

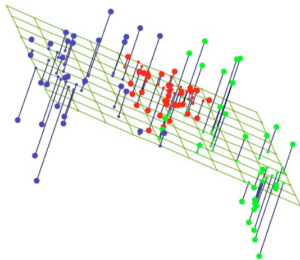
Overview

- Motivation for dimensionality reduction
- Feature selection
 - Wrappers
 - Filters
 - Embedded methods
- Feature transformation
 - Principal Component Analysis (PCA)
 - Autoencoders
 - Clustering

Feature transformation

Linear feature transformation

- $\mathbf{x} \in \mathbb{R}^D \rightarrow \mathbf{y} \in \mathbb{R}^M, D \gg M$
- linear transformation of original space: $\mathbf{y} = \mathbf{U}^T \mathbf{x}, \mathbf{U} \in \mathbb{R}^{D \times M}$



Feature transformation

Linear feature transformation: Example

Assuming an input vector $\mathbf{x} = [x_1, x_2]^T \in \mathbb{R}^{2 \times 1}$ and a transformation $\mathbf{U}\mathbf{x}$, what transformation do each of the following matrices perform?

If $\mathbf{U} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, then $\mathbf{U}\mathbf{x} = \begin{bmatrix} u_{11}x_1 + u_{12}x_2 \\ u_{21}x_1 + u_{22}x_2 \end{bmatrix} \in \mathbb{R}^{2 \times 1}$.

If $\mathbf{U} = \begin{bmatrix} u_{11} & u_{12} \end{bmatrix} \in \mathbb{R}^{1 \times 2}$, then $\mathbf{U}\mathbf{x} = [u_{11}x_1 + u_{12}x_2] \in \mathbb{R}$.

① $\mathbf{U} = [1, 0; 0, 1]$

② $\mathbf{U} = [1.5, 0; 0, 1.5]$

③ $\mathbf{U} = [0, 1; 1, 0]$

④ $\mathbf{U} = [1, 0]$

⑤ $\mathbf{U} = [1, 1]$

⑥ $\mathbf{U} = [1, 1; 0, 1]$

Feature transformation

Linear feature transformation: Example

Assuming an input vector $\mathbf{x} = [x_1, x_2] \in \mathbb{R}^2$ and a transformation $\mathbf{U}\mathbf{x}$, what transformation do each of the following matrices perform?

If $\mathbf{U} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, then $\mathbf{U}\mathbf{x} = \begin{bmatrix} u_{11}x_1 + u_{12}x_2 \\ u_{21}x_1 + u_{22}x_2 \end{bmatrix} \in \mathbb{R}^{2 \times 1}$.

If $\mathbf{U} = \begin{bmatrix} u_{11} & u_{12} \end{bmatrix} \in \mathbb{R}^{1 \times 2}$, then $\mathbf{U}\mathbf{x} = [u_{11}x_1 + u_{12}x_2] \in \mathbb{R}$.

- ① $\mathbf{U} = [1, 0; 0, 1]$ Identity
- ② $\mathbf{U} = [1.5, 0; 0, 1.5]$ Dilation
- ③ $\mathbf{U} = [0, 1; 1, 0]$ Flipping of axes
- ④ $\mathbf{U} = [1, 0]$ Preserving only first dimension
- ⑤ $\mathbf{U} = [1, 1]$ Substituting the first dimension by the sum of the two. Removing the second dimension.
- ⑥ $\mathbf{U} = [1, 1; 0, 1]$ Substituting the first dimension by the sum of the two. Preserving the second dimension.